Detailed C++ Mini Project Presentation

CANTEEN MENU SYSTEM

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Introduction

- The Canteen Menu System is a console-based application developed using C++.
- The project helps users view available food items according to day and time.
- It allows customers to place orders, calculate total bills, and choose payment methods.
- The system demonstrates the use of arrays, loops, conditions, and functions in programming.

Problem Statement

- Managing canteen orders manually can be time-consuming and error-prone.
- Customers may face delays while checking menu items and billing.
- The project aims to automate the process of menu display and bill generation.
- It provides a simple digital solution for canteen management.

Objectives

- To display menu items based on selected day and time.
- To allow users to order multiple items with quantities.
- To calculate the total bill automatically.
- To provide online and offline payment options.
- To improve efficiency in canteen order management.

Software Requirements

- Programming Language: C++
- Compiler: Turbo C++ / Code::Blocks / VS Code
- Operating System: Windows / Linux
- Memory Requirement: Minimal
- User Interface: Console-based

System Description

- The system stores menu items and prices using 3-dimensional arrays.
- Menus are categorized according to days and time slots.
- The user selects a day and preferred time slot.
- The system displays food items and accepts user orders.
- The final bill is calculated and payment mode is selected.

Data Structures Used

- Arrays are used to store menu items and prices.
- 1D Arrays store days and time slots.
- 3D Arrays store food items for each day and time.
- Integer variables are used for billing and quantity calculation.
- Loops are used for menu display and repeated ordering.

Working Procedure

- Step 1: Display available days.
- Step 2: User selects a day.
- Step 3: Display time slots.
- Step 4: User selects a preferred time.
- Step 5: Display corresponding menu items and prices.
- Step 6: Accept item number and quantity.
- Step 7: Calculate total amount.
- Step 8: Repeat ordering if required.
- Step 9: Display final bill and payment options.

Code Explanation

- String arrays are used to store day names and time slots.
- 3D arrays store menu items for every day and time.
- Nested arrays are used to organize food categories efficiently.
- The do-while loop allows repeated ordering.
- Conditional statements handle payment options and validation.

Sample Output

- ===== CANTEEN MENU SYSTEM =====
- 1. Monday 2. Tuesday 3. Wednesday
- Select Time: Morning / Afternoon / Evening
- Menu Displayed with Prices
- User enters item number and quantity
- Final bill amount is generated
- Payment successful message displayed

Advantages

- Reduces manual errors in billing.
- Improves speed of order processing.
- Simple and easy to use.
- Helps organize menu items efficiently.
- Can be implemented in small canteens and cafeterias.

Limitations

- The system is console-based and has no graphical interface.
- Menu items are fixed and stored manually in arrays.
- No database connectivity is used.
- No login or authentication feature.

Future Enhancements

- Add graphical user interface (GUI).
- Connect the system with a database.
- Add online payment gateway integration.
- Include customer login and order history.
- Generate digital receipts automatically.

Conclusion

- The Canteen Menu System successfully automates menu display and billing.
- The project demonstrates practical implementation of C++ concepts.
- It provides a simple and efficient ordering system.
- The system can be further improved with advanced technologies.

THANK YOU